

WORK SHEET - 03

LAW OF MOTION + FRICTION + CIRCULAR

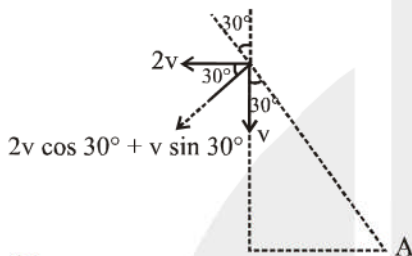
[HINTS & SOLUTIONS]

Section-A

Q.9 (A)

$$\omega_{BA} = \frac{(v_{B/A})_{\perp}}{R} = \frac{2v \cos 30^{\circ} + v \sin 30^{\circ}}{2r}$$

$$= \frac{v\sqrt{3} + \frac{v}{2}}{2r}$$



Q.12 (B)

Choosing the positive X-Y axis as shown in the fig, the momentum of the bead at A is $\vec{p}_i = +m\vec{v}$ The momentum of the bead at B is $\vec{p}_f = -m\vec{v}$

Therefore, the magnitude of the change in momentum between A and B is

$$\Delta \vec{p} = \vec{p}_f - \vec{p}_i = -2m\vec{v}$$

i.e. $\Delta p = 2mv$ along positive X-axis.

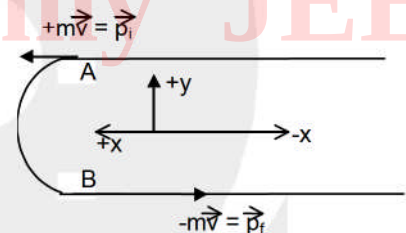
The time interval taken by the bead to reach from A to B is

$$\Delta t = \frac{\pi \cdot d/2}{v} = \frac{\pi d}{2v}$$

Therefore, the average force exerted by the bead on the wire is

$$F_{av} = \frac{\Delta p}{\Delta t} = \frac{2mv}{\frac{\pi d}{2v}} = \frac{4mv^2}{\pi d}$$

Therefore (B).



Q.13 (A)

The ball moves along X-axis with an acceleration, say a , whereas it is in equilibrium along y axis

$$\Rightarrow \Sigma F_x = 0 \Rightarrow R \sin \theta = ma$$

$$\text{and } F_y = 0 \Rightarrow R \cos \theta = mg$$

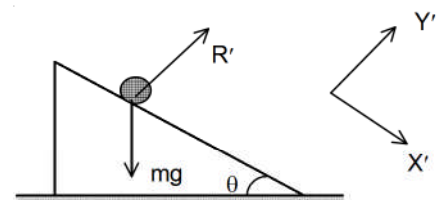
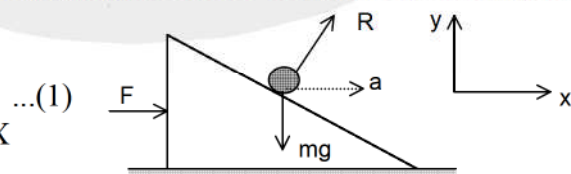
$$\Rightarrow R = mg \sec \theta$$

Here, the ball moves down the plane, along X axis, say with an acceleration a

$$\Rightarrow mg \sin \theta = ma'$$

The ball does not move perpendicular to the plane. It is in equilibrium along y axis

$$\Rightarrow mg \cos \theta - R = 0$$



$$\Rightarrow R' = mg \cos \theta \quad \dots(2)$$

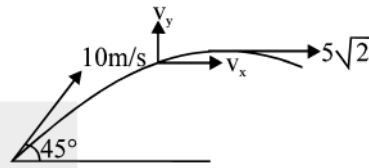
$$(1) \div (2) \Rightarrow \frac{R}{R'} = \sec^2 \theta$$

Therefore (A)

Q.14 (C)

$$v_y = \frac{10}{\sqrt{2}} - 10 \times \frac{1}{\sqrt{2}} = 0$$

$$a_n = \frac{V^2}{R} \Rightarrow R = \frac{(5\sqrt{2})^2}{g} = \frac{50}{10} = 5 \text{ m}$$



Q.15 (D)

|Velocity of B w.r.t. A| = $V_{BA} = 2V$

Now, acceleration of B w.r.t. A = acceleration of B w.r.t. the centre

$$\frac{(2V)^2}{(R_B)_A} = \frac{V^2}{R}$$

$$\therefore (R_B)_A = 4R$$

Q.16 (B)

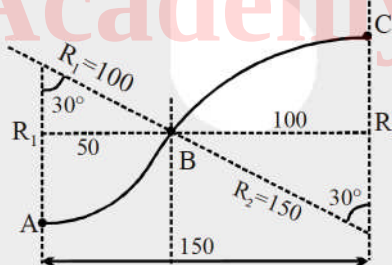
$$S_{AC} = S_{AB} + S_{BC}$$

$$= R_1 \left(\frac{\pi}{6} \right) + R_2 \left(\frac{\pi}{6} \right)$$

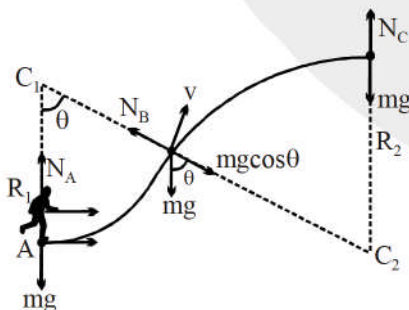
$$= \left(\frac{100}{6} + \frac{200}{6} \right) \pi$$

$$S_{AC} = \frac{300}{6} \pi = 50 \pi \text{ m}$$

$$t = \frac{S_{AC}}{V} = \frac{50\pi}{5} = 10 \pi \text{ s}$$



Q.17 (C)



$$N_A - mg = \frac{mv^2}{R_1}$$

$$N_A = mg + \frac{mv^2}{R_1} = m \left(g + \frac{V^2}{R_1} \right) = m(10.5)$$

$$\Rightarrow (N_A > N_C > N_B) \quad]$$

$$\text{Similarly } N_B = mg \cos \theta = mg \frac{\sqrt{3}}{2} = m(8.66)$$

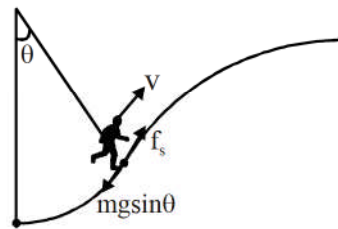
$$\& N_C = mg - \frac{mv^2}{R_2} = m \left(g - \frac{V^2}{R_2} \right) = m(9.75)$$

- Q.18 (D)
Since v is constant

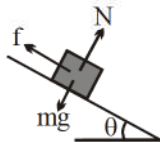
Hence tangential acceleration = 0

$$\Rightarrow f_s = mg \sin \theta$$

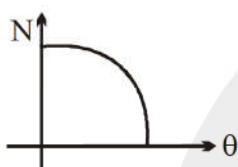
$$\Rightarrow f_A < f_B < f_C$$



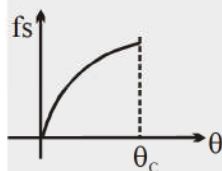
- Q.21 (AB)



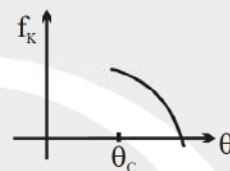
$$N = mg \cos \theta$$



for no slipping
 $f_s = mg \sin \theta$



After slipping
 $f_k = \mu_k N = \mu_k mg \cos \theta$



- Q.22 (ABCD)

As $T \sin 60^\circ = \frac{mv^2}{r}$ (1)

$$T \cos 60^\circ = mg$$
 (2)

$$\therefore \tan 60^\circ = \frac{v^2}{rg}$$

$$v = \sqrt{rg \tan 60^\circ} \quad \text{Also, } r = \ell \sin 60^\circ$$

$$\text{Time period} = \frac{2\pi r}{v} = \frac{2\pi r}{\sqrt{rg \tan 60^\circ}} = 2\pi \sqrt{\frac{r}{g \tan 60^\circ}}$$

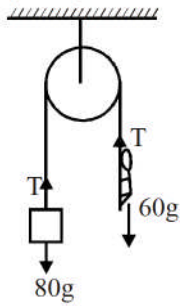
$$= 2\pi \sqrt{\frac{\ell \cos 60^\circ}{g}}$$

From equation (1) $T = 2mg$

$$v = \sqrt{\ell g \sin 60^\circ \tan 60^\circ}$$

$$\therefore \text{Centripetal acceleration } a_c = \frac{v^2}{r} = \frac{rg \tan 60^\circ}{r} = g \tan 60^\circ]$$

Q.24 (ABC)



$$T - 60g = 60a$$

$$900 - 600 = 60a$$

$$a = 5 \text{ m/s}^2$$

$$T - 80g = 80a$$

$$a_{\text{block}} = \frac{5}{4} \text{ m/s}^2$$

$$V_{\text{man}} = at = 10 \text{ m/s}$$

$$V_{\text{block}} = a_b \times t = \frac{5}{2} \text{ m/s}$$

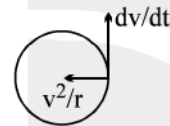
Both are moving, so string becomes slack
acceleration = $g \downarrow$

Q.26 $a = \frac{4mg}{M+17m}$

Q.27 $[500\sqrt{5}]$

$$F_{\text{net}} = m \sqrt{\left(\frac{v^2}{r}\right)^2 + \left(\frac{dv}{dt}\right)^2} = m \sqrt{\left(\frac{1}{2}\right)^2 + (1)^2}$$

$$= \frac{m}{2} \sqrt{5} = 500\sqrt{5} \text{ N Ans.}$$



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